

READER

Lesson 1 — The workbench as a test lab

AI agents on the workbench

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Introduction

KEY POINTS

- An AI agent can run the entire test cycle on its own.

1.1 The workbench as a test lab

The workbench as a test lab

AI agents that flash, measure and test on their own



In this lesson we build, step by step, a setup in which an AI agent can flash and test firmware on its own.

1.2 The AI agent can:

THE AI AGENT CAN:

- Flash firmware
- Watch the serial log
- Run tests
- Validate the results
- Fix problems in the code

→ Your workbench becomes an automated test lab



The agent uses ordinary command-line tools such as `esptool` and a serial monitor. See also the [ESP-IDF documentation](#).

1.3 Let the machine do the boring work, and keep the overview...

“

Let the machine do the boring work, and
keep the overview yourself.



Questions

1. What does the agent do when a test fails?

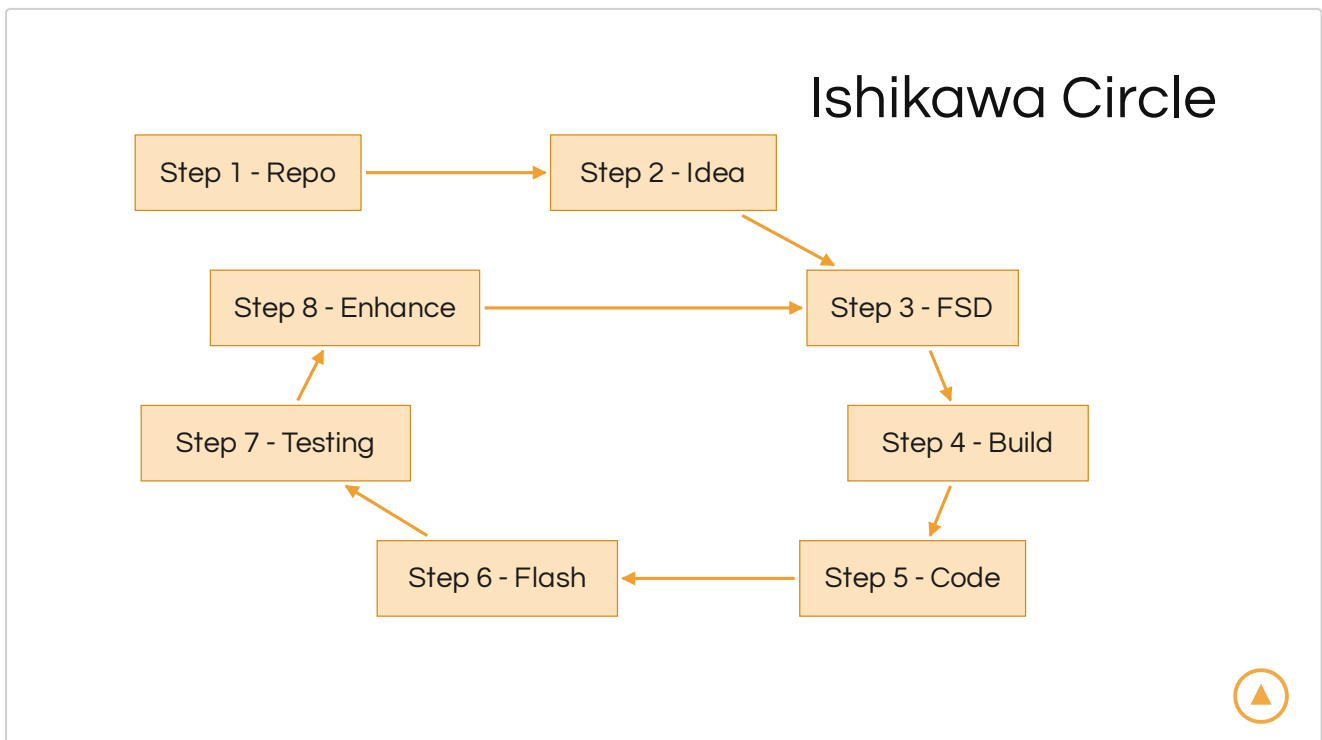
- ☐ It stops and waits for the developer
- ☐ It adjusts the code and starts over
- ☐ It flashes the same firmware again

Workflow

KEY POINTS

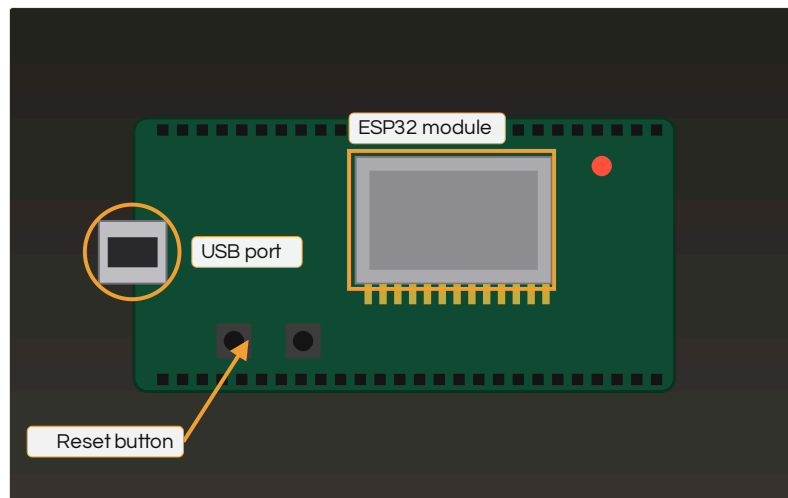
- Development is a cycle, not a straight line.
- Know the main components of your board.

2.1 Ishikawa Circle



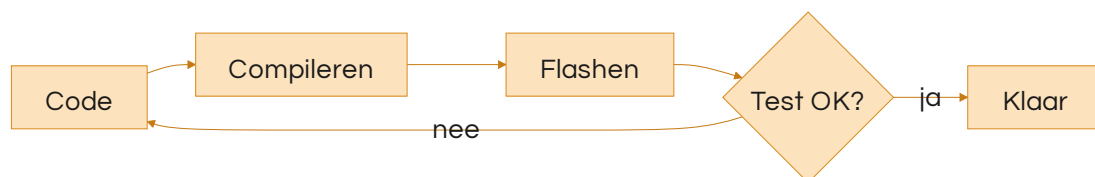
2.2 The board

THE BOARD



2.3 The test cycle

THE TEST CYCLE



2.4 A test in Python

A TEST IN PYTHON

```
import serial
port = serial.Serial("/dev/ttyUSB0", 115200, timeout=10)

def wait_for(text):
    line = port.read_until(text.encode())
    return text in line.decode()

assert wait_for("BOOT OK")
```



2.5 Baud rate and time

BAUD RATE AND TIME

$$t_{byte} = \frac{10}{\text{baud rate}}$$

At 115200 baud, one byte takes about $87 \mu s$.



2.6 Manual or with an agent?

MANUAL OR WITH AN AGENT?

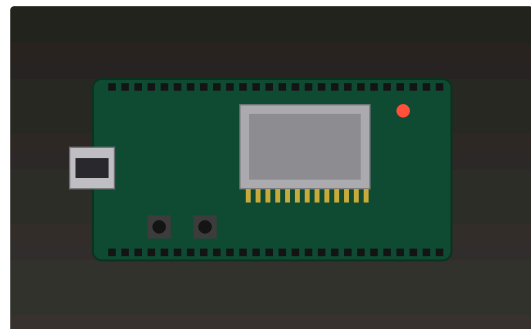
Task	Manual	With agent
Flashing	1 min	automatic
Reading logs	eyes on screen	continuous
Testing	when you remember	every build



2.7 Setup

SETUP

- Laptop with agent
- USB hub
- Test board



2.8 Thank you!

Thank you!

Questions? Comments? Let me know.

Questions

1. What happens after step 8, Improve?

- ☐ The project is finished
- ☐ The circle starts again at the functional design
- ☐ We start over with an empty repository

2. Which component carries the power supply and the serial connection?

- ☐ The USB port
- ☐ The reset button
- ☐ The ESP32 module

Answers

Chapter 1 — Introduction

1. It adjusts the code and starts over

Chapter 2 — Workflow

1. The circle starts again at the functional design
2. The USB port

Sources

1. ESP-IDF documentation

<https://docs.espressif.com/>